

Yunjae Kim

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EDUCATION

St. Thomas Aquinas Regional Secondary School

North Vancouver, Canada

Dogwood Diploma; GPA: 4.0/4.0 (Unweighted)

Sep. 2022 – Present

- SAT: 1550/1600 (750/800 Reading & Writing; 800/800 Math)
- Completed 11 university-level AP exams: Calculus BC, Physics C (Mechanics & Electricity and Magnetism), Physics 1 & 2, Chemistry, Computer Science A, Statistics, English Language and Composition, Psychology, World History.

RESEARCH EXPERIENCE

Student Researcher

Oct. 2025 – Present

The University of British Columbia (Zhou Group)

Vancouver, Canada

- Investigated lattice anharmonicity and its relationship to potential wells to predict high-temperature material properties.
- Modeled interatomic interactions using a Morse potential to derive an analytical expression for the anharmonic correction to heat capacity.
- Developed a Python nonlinear regression model using thermodynamic data to analyze how bond strength affects heat capacity deviations in crystalline solids.

Quantum Leaps Fellow

Nov. 2025 – Present

British Columbia Institute of Technology

Burnaby, Canada

- Analyzed the temperature dependence in MOSFET performance at cryogenic temperatures.
- Applied transport theory and phonon physics to predict cryogenic MOSFET behaviour and planned experimental procedures.
- Awarded \$ 1000 in grants for the project by the Society for Canadian Women in Science and Technology.

Summer Research Student

Jul. 2026

Pritzker School of Molecular Engineering, University of Chicago

Chicago, USA

- Fabricated, characterized, and optimized electronic transition and interfacial charge transport phenomena in dye-sensitized nanocrystalline TiO₂ solar cells
- Developed a novel modification of Coumarin 6 dye with a selenophene π -conjugating group and cyano/phosphonic acid electron-accepting groups and performed computational analysis using time-dependent DFT.

PUBLICATIONS

Y. Kim, “Analytically Derived Symmetric VSEPR Geometries from Classical Energy Minimization on a Sphere,” *Chem. Educator* (2026) 31(1): 3082738, doi.org/10.1525/tce.2026.3082738.

Y. Kim, “Fabrication and Characterization of an Anthocyanin-Sensitized TiO₂ Solar Cell,” *ChemRxiv* (2026). doi.org/10.26434/chemrxiv.15006409/v1

Y. Kim, “The Language of Symmetry: Group Theory and Its Applications to Physics,” *J. Young Phys.* (2026). <https://www.journalofyoungphysicists.org/post/the-language-of-symmetry-group-theory-and-its-applications-to-physics>

SKILLS

Experimental: Thin-film deposition, spin-coating, FTO substrate prep, UV-Vis spectroscopy, I – V characterization, titration, ELISA

Computational/Software: Python (Pandas, NumPy, SciPy, Matplotlib), Molecular simulation (DFT/TD-DFT), linear/non-linear regression, numerical methods, Java, R (figure generation), LaTeX

Professional Skills: Scientific writing & presentation, grant proposals, STEM mentoring, interdisciplinary collaboration

PRESENTATIONS

Y. Kim. *DFT Study of [2-cyano-2-[5-[7-(diethylamino)-2-oxo-chromen-3-yl]selenophen-2-yl]ethenyl]phosphonic acid (DEAC-Se-P) as a Novel Coumarin 6 Derivative*, Oral Presentation at the Pathways in Molecular Engineering Symposium, Chicago, US, 23 Jul 2026.

ENRICHMENT

GCI World

Sep. 2026 - Current

The University of Tokyo (Matsuo-Iwasawa Lab)

Tokyo, Japan (Virtual)

- Selected to study the mathematical and computational foundations of data science with applications of machine learning.

Quantum School for Young Students

Jul. 2026 - Aug. 2026

Institute for Quantum Computing, University of Waterloo

Waterloo, Canada (Virtual)

- Selected 1 of 200 nationally to attend the 2026 Quantum School for Young Students (QSYS).
- Studied the mathematical formalism of quantum mechanics, quantum computing, algorithms, cryptography, and optics.

GoPhysics! Gravity & Black Holes Workshop

Jul. 2026

Perimeter Institute for Theoretical Physics

Waterloo, Canada (Virtual)

- Selected by a world-leading institute for theoretical physics to attend the GoPhysics! workshop.
- Studied Einstein's theory of general relativity, concepts of differential geometry, and black holes.

CERN-Solvay Education Programme

May. 2026

European Organization for Nuclear Research

Geneva, Switzerland (Virtual)

- Participated in the online portion of the program.
- Completed modules on group theory and the Standard Model, thermodynamics, and phase transitions.

LEADERSHIP

STA Math Club | President

Sep. 2025 – Present

- Lead club operations and teams, focusing on advanced problem-solving.
- Coordinate mock competitions to build performance confidence and analytical skills.
- Results: Qualifying the grade 9 team for the Math Challengers Provincial Finals; placing the school in 10th place on the team honor roll for the Pascal Math Contest.

STA Computer Science Club | Founder & President

Sep. 2024 – Present

- Develop curriculum focused on Java fundamentals, mentor students for the Canadian Computing Competition.
- Manage all club logistics including scheduling, membership, and project coordination.

Simon Fraser University | Camp Volunteer

Aug. 2026

- Summer volunteer for the chemistry and science junior camps.
- Introduce various chemical topics such as forensic science, states of matter, chemical kinetics and enzymatic activity.
- Introduce various aspects of Biology, Chemistry, Engineering, and Earth Science.

The University of British Columbia | Teaching Assistant

Jul. 2025

- Teaching assistant at the Department of Physics and Astronomy's Phenomenal Physics Summer Camp.
- Assisted instructors in facilitating daily hands-on physics activities and demonstrations for young campers.
- Introduced topics including gravity, energy, momentum, buoyancy, and electricity.